



Unit 3 · Outcome 1 · Maintaining Biodiversity

# Sustainability principles for biodiversity conservation

**KK15** VCE Environmental Science · Units 3 & 4 · Exam study summary

**Key knowledge.** · KK15 Every other dot point in this chapter described what biodiversity is and how we protect it

Digital Vector EnviroSci · single-topic mastery summary with worked good-vs-bad answers, exam traps, command words and practice.



## The decision lens

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01 The decision lens Key knowledge · KK15 Every other dot point in this chapter described what biodiversity is and how we protect it. This one is different. It gives you a framework for judging whether a management decision is actually sustainable — not just whether it works today, but whether it is fair, efficient and safe for the long run. The VCAA study design names six sustainability principles that apply, word for word, across Outcomes on biodiversity, sustainable development, climate change and energy. Learn them once here and you carry them through

## Our running case: Kestrel Ridge Reserve

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02 Our running case: Kestrel Ridge Reserve ▲ The scenario we'll evaluate Kestrel Ridge Reserve is a fragmented patch of swamp-gum and manna-gum woodland on the fringe of Melbourne — the last stronghold of a critically endangered endemic bird, the Helmeted Honeyeater ( *Lichenostomus melanops cassidix* ), Victoria's bird emblem. Fewer than 250 mature birds survive in the wild. A management committee must choose between competing proposals: a captive-breeding-and-release program, buying back cleared farmland to widen a habitat corridor, or doing nothing and

## The six sustainability principles

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03 The six sustainability principles Learn the exact VCAA wording These are the exact six from the study design. Memorise the names in this order — examiners use these labels and reward the precise terms. Each card gives the definition and how it applies to maintaining biodiversity. △ Two pairs that get mixed up Inter generational = between generations (you and your grandchildren). Intra generational = within one generation (people alive now, e.g. rich vs poor nations). The prefix is the whole answer — "inter" (think international , across) vs "intra" (t

## See intergenerational equity in motion

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04 See intergenerational equity in motion Intergenerational equity is the hardest principle to feel , because the people it protects aren't born yet. Drag the slider to advance the generations at Kestrel Ridge, and switch between two management paths to watch how today's choice ripples forward to 2125. 3D timeline unavailable — use the slider and readout below; all content still works. Management path Protect habitat now Clear for short-term gain Generation 2025 · today Wild population ~230 birds      Analogy · the inheritance, not the windfall Intergenerat

## The precautionary principle & competing pressures

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05 The precautionary principle & competing pressures Real decisions rarely satisfy all six principles at once — they pull against each other. The precautionary principle settles the most dangerous tension of all: what to do when the science is uncertain but the harm could be irreversible. Tilt the balance and toggle the uncertainty to see why caution wins when extinction is on the table. 3D balance unavailable — the controls and verdict below still work. Decision weight ◀ Short-term economic gain Ecological integrity ▶ Outcome is uncertain & possibly irr

## Applying all six to the corridor buy-back

06 Applying all six to the corridor buy-back Here's the payoff. The committee chooses to buy back farmland to widen the Kestrel Ridge corridor and run a small captive-breeding program . This single decision can be evaluated against every principle — exactly what an "evaluate" question wants: Principle How the corridor buy-back measures up Biodiversity & ecological integrity Widening the corridor reconnects fragmented patches, restoring the woodland's structure and food web so it can self-sustain — not just the bird, but the whole community it lives in. E

| Principle                           | How the corridor buy-back measures up                                                                                                                                                 |
|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Biodiversity & ecological integrity | Widening the corridor reconnects fragmented patches, restoring the woodland's structure and food web so it can self-sustain — not just the bird, but the whole community it lives in. |
| Efficiency of resource use          | Targeting the cheapest land that yields the biggest connectivity gain achieves the goal with the least money and area. Captive breeding is sized to need, avoiding waste.             |
| Intergenerational equity            | Securing habitat now means future Victorians inherit a living species and its ecosystem services, rather than an extinct emblem in a museum.                                          |
| Intragenerational equity            | Fair only if affected landholders are compensated and traditional owners share in decisions and benefits — costs must not fall on one group while others gain.                        |
| Precautionary principle             | Acting before population models are certain is justified because the risk (extinction) is serious and irreversible.                                                                   |
| User pays principle                 | If a developer's clearing caused the fragmentation, requiring them to fund offsets/restoration internalises the environmental cost rather than passing it to the public.              |

## Sort the scenarios

### ◇ ACTIVE RECALL

07 Sort the scenarios Drag · 9 cards Each card is a snippet of reasoning from a real conservation decision. Drag it onto the principle it best illustrates. Get instant feedback — and the running tally feeds your score dock. Which principle is this? Placed correctly: 0 / 9 ↺ Reveal answers

## Decode the command word

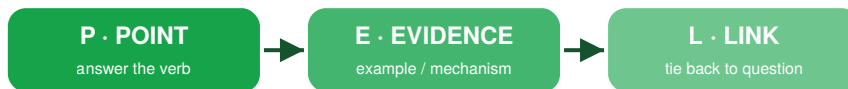
### ◆ COMMAND WORDS

08 Decode the command word The verb in the question tells you exactly how much to write. Tap each to see what the examiner is really asking for on a sustainability-principles question.

## P·E·L writing trainer

### ✎ P·E·L WRITING

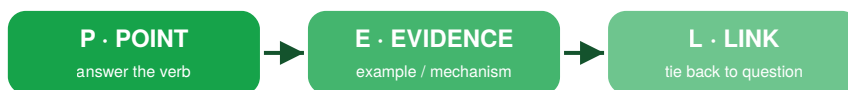
10 P·E·L writing trainer Point · Explain · Link The single most reliable structure for principle questions. P oint — name the principle. E xplain — define it in VCAA terms. L ink — tie it to the specific scenario evidence. Try it on the prompt below, then reveal a model. Prompt A council rejects a housing development next to Kestrel Ridge because it might disturb the Helmeted Honeyeater, though no study has yet confirmed harm. Explain which sustainability principle most supports the council's decision. (3 marks) P Point — name the principle State it prec



The P·E·L answer structure — make a Point, support with Evidence, then Link to the question.

## Good vs bad answer, marked

11 Good vs bad answer, marked Question: "A mining company is required to pay for the rehabilitation and replanting of land it cleared near a threatened-species habitat. Identify and explain the sustainability principle this reflects." (3 marks) X Weak response 1 / 3 "This is sustainable because the company is helping the environment by replanting . It's a good thing to do and shows they care about protecting nature for the future." Names no principle. "Helping the environment" and "good thing to do" are not VCAA terms. No definition, no link to who pays



The P·E·L answer structure — make a Point, support with Evidence, then Link to the question.

## Practice questions

### ★ PRACTICE & SELF-MARK

12 Practice questions 5 questions · 20 marks Exam-style questions building from short-answer to full evaluation. Write your answer first, then reveal the model, marking guide and a worked weak response. Mark yourself honestly — your scores collect in the dock.

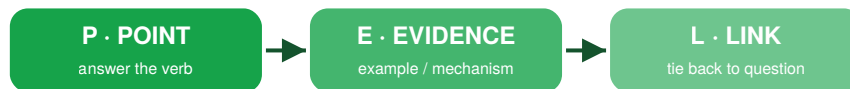
## 60 -second recap

### ♦ RECAP / CHEAT SHEET

13 60-second recap ♦ If you remember nothing else ① There are six principles: biodiversity & ecological integrity, efficiency of resource use, intergenerational equity, intragenerational equity, the precautionary principle, and the user pays principle. ② Inter = across generations; intra = within one generation. ③ The precautionary principle = act under uncertainty when harm is serious/irreversible. ④ User pays = the polluter funds the cost. ⑤ To evaluate , weigh principles upheld against principles compromised, then judge. ⑥ Always name → define → link

## ★ Good vs bad worked answers

Study the contrast — the same question answered two ways. The strong column shows the moves that earn the marks.



The P-E-L answer structure — make a Point, support with Evidence, then Link to the question.

### ✗ WEAK ANSWER

This is sustainable because the company is helping the environment by replanting . It's a good thing to do and shows they care about protecting nature for the future." Names no principle. "Helping the environment" and "good thing to do" are not VCAA terms. No definition, no link to who pays for the cost . Scores 1 mark at most for a vague gesture at sustainability.

### ✓ STRONG / MODEL ANSWER

This reflects the user pays principle . It states that those who use or damage a resource should bear the full cost of that use, including the environmental cost, rather than passing it to the wider community. Here the mining company caused the habitat damage, so requiring it to fund rehabilitation internalises the cost of clearing rather than leaving taxpayers or future generations to pay. " Names the principle, defines it in VCAA language, and links it to the specific evidence (the company caused the damage, so it pays ). Full marks.

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Worked good/weak answers illustrate exam technique — always check against current VCAA marking advice.